# Torque and voltage $dq$ identities (Issue #68)

Sensitivity to PointPer and mesh size

## 1 Identity checks and setup

For two 5-phase SynRM models (40 slots and 60 slots), we test the *amplitude-invariant* Park identities against Ansys Maxwell 2D:

$$T_{dq} = \frac{m}{2}\, p\, \left[(\psi_{d1} i_{q1} - \psi_{q1} i_{d1}) + 3(\psi_{d3} i_{q3} - \psi_{q3} i_{d3})\right], \tag{1}$$

$$V_{q1} - R_s I_{q1} = \omega \psi_{d1}, \qquad V_{d1} - R_s I_{d1} = -\omega \psi_{q1}. \tag{2}$$

The third-harmonic voltage checks use subscript 3 and $3\omega$; $m = 5$ (phases), $p$ is the pole-pair count, and $\omega = 2\pi f$. For each check, $A$ and $B$ denote the period means of the two compared quantities (the left and right sides for voltage, or $T_{dq}$ and `Moving1.Torque` for torque). The relative error is

$$\varepsilon_{\text{rel}} = \frac{|A - B|}{|B|} \times 100\%. \tag{3}$$

Period means use `np.mean`, the discussion criterion is 5%.

The tests use $f = 50\,\text{Hz}$, $N_{\text{per}} = 1$, fixed currents $(i_{d1}, i_{q1}, i_{d3}, i_{q3}) = (0.8, 0.9, 0.2, -0.15)\,\text{A}$ and the OneLambda rotor (seed 42). The time step is $\Delta t = 1/[f(\text{PointPer} - 1)]$. The 3 mm reference mesh is compared with a 0.5 mm maximum element size applied to the *stator, rotor, air gap and coils*.

## 2 Sensitivity results

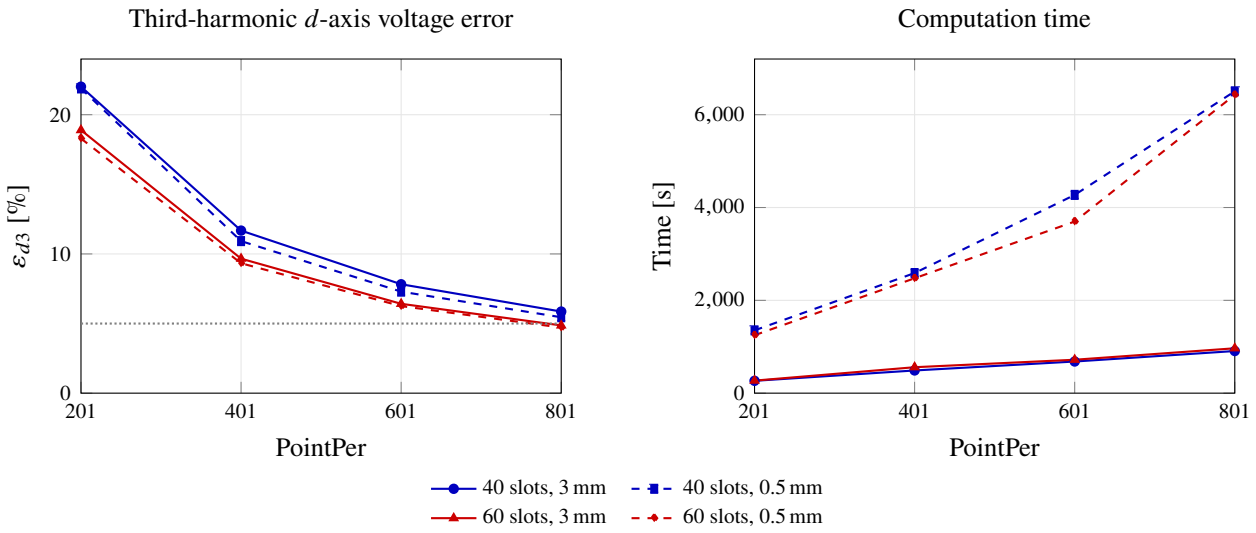



Figure 1: Mesh comparison for the common PointPer range (201–801). The dotted line marks the 5% error threshold.

**PointPer matters more.** At 3 mm, $\varepsilon_{d3}$ drops from 22.02% to 5.86% (40 slots) and from 18.89% to 4.87% (60 slots) as PointPer increases from 201 to 801. All five errors on the 3 mm mesh first fall below 5% at PointPer 1001 (40 slots) or 801 (60 slots).

**Finer meshing offers little improvement.** At PointPer 801, refining from 3 to 0.5 mm reduces $\varepsilon_{d3}$ only from 5.862% to 5.453% (40 slots) and from 4.866% to 4.712% (60 slots), but raises runtime by 7.2 times and 6.6 times.

**Conclusion.** Increasing PointPer is more effective than refining the mesh for these $dq$ checks. When criterion is set to 5%, the values of PointPer are 1001 for 40 slots and 801 for 60 slots, respectively.

## 3 Complete sweep data

All 21 cases' raw data are listed below. $P$ = PointPer; $h_{\max}$ = maximum mesh length [mm]; $t$ = elapsed computation time [s]. $\varepsilon_T$ is the torque-identity error; $\varepsilon_{q1}$, $\varepsilon_{d1}$, $\varepsilon_{q3}$ and $\varepsilon_{d3}$ are the first- and third-harmonic $q$-/$d$-axis voltage identity errors. All five errors are in percent.

Table 1: Tested cases (errors in %; bold rows: all five errors below 5%).

| | | | | Relative error [%] | | | | |
|---|---|---|---|---|---|---|---|---|
| Slots | $P$ | $h_{\max}$ [mm] | $t$ [s] | $\varepsilon_T$ | $\varepsilon_{q1}$ | $\varepsilon_{d1}$ | $\varepsilon_{q3}$ | $\varepsilon_{d3}$ |
| 40 | 201 | 3 | 266 | 0.1452 | 0.2168 | 8.8363 | 0.9687 | 22.0154 |
| 40 | 401 | 3 | 489 | 0.3381 | 0.1070 | 4.7957 | 0.4139 | 11.6775 |
| 40 | 601 | 3 | 682 | 0.3864 | 0.0781 | 3.1646 | 0.3021 | 7.8227 |
| 40 | 801 | 3 | 908 | 0.4118 | 0.0593 | 2.3503 | 0.2384 | 5.8617 |
| **40** | **1001** | **3** | **1981** | **0.4164** | **0.0479** | **1.8624** | **0.1937** | **4.6725** |
| **40** | **1201** | **3** | **1370** | **0.4206** | **0.0400** | **1.5430** | **0.1625** | **3.8838** |
| **40** | **2001** | **3** | **4169** | **0.4300** | **0.0241** | **0.9144** | **0.0980** | **2.3215** |
| 60 | 201 | 3 | 270 | 0.2944 | 0.3223 | 9.1336 | 0.8636 | 18.8941 |
| 60 | 401 | 3 | 559 | 0.1740 | 0.1391 | 4.1977 | 0.4237 | 9.6603 |
| 60 | 601 | 3 | 720 | 0.1738 | 0.0889 | 2.7516 | 0.2925 | 6.4179 |
| **60** | **801** | **3** | **967** | **0.2028** | **0.0737** | **2.1236** | **0.2492** | **4.8662** |
| **60** | **1201** | **3** | **2003** | **0.2221** | **0.0531** | **1.4476** | **0.1812** | **3.2774** |
| **60** | **2001** | **3** | **76474** | **0.2313** | **0.0322** | **0.8652** | **0.1090** | **1.9687** |
| 40 | 201 | 0.5 | 1353 | 0.2507 | 0.2074 | 8.8112 | 0.9326 | 21.8884 |
| 40 | 401 | 0.5 | 2588 | 0.0076 | 0.1114 | 4.3557 | 0.4760 | 10.9223 |
| 40 | 601 | 0.5 | 4272 | 0.0171 | 0.0751 | 2.9180 | 0.3214 | 7.2923 |
| 40 | 801 | 0.5 | 6504 | 0.0270 | 0.0571 | 2.1867 | 0.2425 | 5.4529 |
| 60 | 201 | 0.5 | 1248 | 0.1930 | 0.3048 | 9.3010 | 0.8571 | 18.2915 |
| 60 | 401 | 0.5 | 2476 | 0.1574 | 0.1594 | 4.4569 | 0.5135 | 9.3301 |
| 60 | 601 | 0.5 | 3698 | 0.1231 | 0.1067 | 2.9596 | 0.3531 | 6.2536 |
| **60** | **801** | **0.5** | **6428** | **0.1031** | **0.0810** | **2.2027** | **0.2642** | **4.7123** |